

Verifying Inverses & Writing Inverses Using Transformations

Name: _____ Period: _____

Part 1: Verifying Inverses by Composition

Two functions f and g are inverses if and only if **both**:

$$f(g(x)) = x$$

AND

$$g(f(x)) = x$$

What this means: If you compose a function with its inverse (in either order), you get back x .

Example 1: Verify that $f(x) = 2x + 6$ and $g(x) = \frac{x-6}{2}$ are inverses.

Check $f(g(x))$:

Check $g(f(x))$:

Since $f(g(x)) = \underline{\hspace{2cm}}$ and $g(f(x)) = \underline{\hspace{2cm}}$, they inverses.

Example 2: Verify that $f(x) = x^3 - 4$ and $g(x) = \sqrt[3]{x+4}$ are inverses.

Check $f(g(x))$:

Check $g(f(x))$:

Since $f(g(x)) = \underline{\hspace{2cm}}$ and $g(f(x)) = \underline{\hspace{2cm}}$, they inverses.

Part 2: Writing Inverses Using Transformations

When you know the transformations, you can write the inverse by **reversing the operations**:

Original Transformation	Inverse Transformation
Vertical shift up k	Horizontal shift right k
Vertical shift down k	Horizontal shift left k
Horizontal shift right h	Vertical shift up h
Horizontal shift left h	Vertical shift down h
Vertical stretch by a	Vertical compression by $\frac{1}{a}$ (then cube/cube root swap)
Reflect over x -axis	Reflect over x -axis

Key Idea: Vertical $\leftrightarrow$ Horizontal swap, and operations reverse!

Example 3: Write the inverse of $f(x) = (x-3)^3 + 5$ using transformations.

Parameter	Original Transformation	Inverse Transformation
$h = 3$	Right 3	
$k = 5$	Up 5	

$$f^{-1}(x) = \underline{\hspace{4cm}}$$

Example 4: Write the inverse of $f(x) = 2\sqrt[3]{x+4} - 1$ using transformations.

Parameter	Original Transformation	Inverse Transformation
$a = 2$	Vertical stretch by 2	
$h = -4$	Left 4	
$k = -1$	Down 1	

$$f^{-1}(x) = \underline{\hspace{4cm}}$$

Example 5: Write the inverse of $f(x) = -(x+2)^3 + 7$ using transformations.

Parameter	Original Transformation	Inverse Transformation
$a = -1$	Reflect over x -axis	
$h = -2$	Left 2	
$k = 7$	Up 7	

$$f^{-1}(x) = \underline{\hspace{4cm}}$$